

A straight line in many rooms

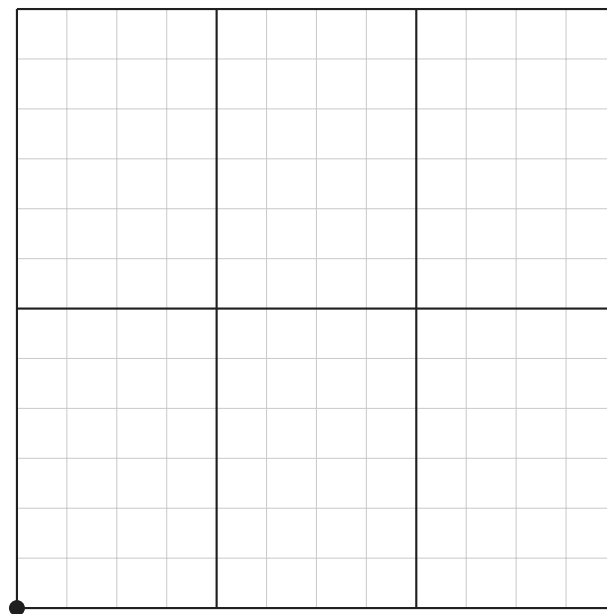
Name: _____

Date: _____

Table sizes are width w by height h . Start bottom-left at 45 degrees. Reflect at walls; stop at the FIRST corner. Count a wall bounce, but not the starting or ending corner.

1. Trace, then unfold.

Find the route on a 4×6 table. If this trace is familiar, begin with the copied rooms and explain your prediction. Mirror copies let a bounce become a straight continuation. On the copied rooms below, draw the diagonal to the first meeting of a vertical and horizontal room wall.



each room is width 4, height 6

2. Name the meeting distance.

Call that horizontal distance L . Why must L be a multiple of *both* w and h ? Why do we take the *least* positive common multiple?

Why a corner means a simultaneous wall meeting:

Try it with objects. Explain by pointing, talking, or drawing.

Predict every ending corner

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3. Fold the rooms back.

If the straight line crosses $m = L/w$ whole room widths, where does it end: right or left? If it crosses $n = L/h$ whole room heights, where does it end: top or bottom?

w	h	L	m	n	Corner
4	6				
6	9				
6	10				
8	14				

4. Two even sides: is that enough?

Compare 4×6 with 6×10 . Both have even side lengths. Why can that rule give the wrong answer?

5. A corner too soon.

Suppose m, n shared a divisor $d > 1$. How many whole widths and heights would the path have crossed at distance L/d ? Why would that already be a corner? Use this to rule out a first ending at bottom-left, which needs both counts even.

An earlier corner would mean:

6. Optional: shrink the side lengths.

With an adult, write $w = ga, h = gb$, where g is their greatest common divisor. The coprime rule says: if a, b share no factor greater than 1, b divides at only when it divides t . For $a = 2, b = 3$, explain this using $3t - 2t = t$. Ask for the general argument. Now $L = gat$ must also be a multiple of gb . Why is the least possible t equal to b ? Conclude $L = gab, m = b, n = a$.

The corner rule in reduced side lengths a and b :

Try it with objects. Explain by pointing, talking, or drawing.

How many bounces is best?

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7. Count walls instead of drawing.

Before the final corner, how many vertical room walls does the unfolded line cross? How many horizontal room walls? Could a meeting be counted twice before the first corner?

A formula in m,n (or a,b after task 6), with a reason:

8. Optional proof continuation: inverse design.

Design a whole-number rectangle ending at the **top-right after exactly four bounces**. Find all the possible reduced width/height pairs. Then make a second size with the same shape.

All reduced pairs, and why there are no others:

9. An impossible order?

An adult requests a table ending at the top-right after exactly **three** bounces. Find one, or prove it cannot exist.

My construction or obstruction:

Researcher's habit: Distinguish a table you failed to find from a table you can prove does not exist.

Try it with objects. Explain by pointing, talking, or drawing.